

SHRIVATHS SHYAM

San Diego, CA

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SKILLS

Programming Languages: Python, JavaScript, SQL, HTML/CSS, Java, C/C++

Frameworks/Libraries: FastAPI, PyTorch, Pandas, LangChain, Numpy, ReactJS, Scikit-learn, Plotly

Cloud/DevOps/Database/OS: Azure, AWS, Terraform, Jenkins, Git, PM2, NGINX, Apache NiFi, Postman, MS SQL Server, Windows, Linux

EDUCATION

University of California, San Diego

Sep 2022 – Jun 2024

MS ECE - Machine Learning and Data Science, GPA: 3.52/4

La Jolla, CA, United States

- **Coursework:** Statistical Learning I, Linear Algebra, ML for Physical Applications, Deep Generative Models, Optimization and Acceleration of DL on Various Hardware platforms, Recommender Systems, Software Engineering, Data Systems for Machine Learning

SSN College of Engineering, Anna University, Chennai

Jun 2016 – Apr 2020

BE - Electronics and Communication Engineering, GPA: 8.88/10

Chennai, TN, India

EXPERIENCE

NLitenData LLC

Aug 2024 – Present

AI Engineer

San Diego, CA, United States

- Designed and developed scalable microservices for product features using **Python (FastAPI)** and **Node.js**.
- Built an **AI agent** using **LangChain** to monitor and classify incoming emails and extracting details, reducing manual processing time.
- Developed a FastAPI **chatbot** using **GPT-4o API** with context-aware retrieval from databases, enhancing user experience.
- Built a **document parsing system** using GPT-4o API and **prompt engineering** to extract structured insights for business workflows.
- Implemented ETL pipelines using **Apache NiFi** to ingest and transform data from external sources into **Azure SQL Database**.
- Designed optimized **database schemas** for application performance, improving efficiency of downstream analytics.
- Performed **end-to-end testing** of APIs and integrations using **Postman**, improving reliability of deployed features.
- Implemented **Infrastructure as Code (IaC)** using **Terraform** to automate provisioning and management of Azure resources.
- Deployed applications on **Azure VM** with **NGINX** and **PM2**, **Git-based CI/CD** for version control.
- Integrated **Cloudflare** for application security, DNS management, traffic optimization, reducing latency and mitigating DDoS threats.
- Led **Scrum sprints** from planning through deployment, promoting collaborative problem-solving across teams.

Talmo Lab, Salk Institute for Biological Studies

Jul 2023 – Jun 2024

Research Engineer

La Jolla, CA, United States

- Built new features for **SLEAP**, an **object detection and pose estimation Python library** with more than 100k downloads.
- Designed and executed **data preprocessing pipelines** (frame extraction, normalization, annotation) using **OpenCV** and **FFmpeg**.
- Trained and optimized **deep learning models (UNet, ResNet)** with **hyperparameter tuning**, improving prediction accuracy.
- Enhanced existing **tracking algorithms** (optical flow, Kalman filtering) to enhance **inference performance**.
- Built modular **APIs** for **data processing, pose estimation, and visualization**, improving usability for researchers.
- Automated packaging and deployment across **Linux, Windows, and macOS** with **GitHub Actions CI/CD**, improving reproducibility.
- Resolved bugs and provided user support via **GitHub issues**.

Cognizant Technology Solutions

Nov 2020 – May 2022

Programmer Analyst

Chennai, TN, India

- Implemented core business functionalities using **Pega-PRPC**, a business process management (BPM) tool, for **Prudential Financial**.
- Designed **REST API flows** and **SQL logic** to streamline data handling in case management systems.
- Developed and hosted **full-stack web services** using **Java, JavaScript**, and **HTML/CSS** via **Apache server**.
- Managed and oversaw the **Continuous Integration/Continuous Deployment (CI/CD)** pipeline using **Jenkins**.

PROJECTS

Fediblend - Fediverse Feed Amalgamator | *HTML, CSS, JavaScript, Git, Agile* | [\[Code\]](#)

Oct 2023 – Dec 2023

- Designed and developed a landing page that pulls trending posts from user-specified instances of the federated social media network.
- Improved the loading time of the page by 30% with parallelization of API calls.
- Implemented test functions in JavaScript with 95% code coverage which ensured that the webpage was robust.

Multi Modal Recommender Systems for Rating Prediction | *Pandas, Vaex, Plotly, PyTorch*

Oct 2023 – Dec 2023

- Performed web scraping of Google Local Reviews data and performed exploratory data analysis (EDA) using Vaex.
- Generated text embeddings with pre-trained **BERT** and image embeddings with pre-trained **VGG16** as multi modal features.
- Developed an Explicit Feedback Recommender System using Deep and Cross Networks, achieving **0.59 MSE** through ablation studies.

Uber Data Analysis Using BigQuery | *SQL, Google Cloud Storage, Mage-AI, Python*

Jul 2023 - Aug 2023

- Designed and executed ETL pipelines to ingest and process Uber trip data using BigQuery and Mage-AI.
- Conducted exploratory data analysis (EDA) and built interactive dashboards for trip pattern analysis.
- Used SQL and Pandas to generate insights into ride demand and pricing trends.

Efficient and Goal-Conditioned Learning in RL using TD-MPC | *PyTorch, Python, Linux, Git* | [\[Code\]](#)

Apr 2023 – Jun 2023

- Improved the performance of existing Temporal Difference-Model Predictive Control (**TD-MPC**) by using offline demos.
- Experimented with conditioned goal generation using a GAN and achieved faster goal completion with 25% lesser training steps (~150k steps) compared to fixed goal (~200k steps).

Knowledge Distillation on Transformers | *Python, PyTorch, Transformers*

Apr 2023 – Jun 2023

- Performed fine-tuning of pre-trained BERT and GPT-2 models using textbrewer on SST-2 and CoLA datasets.
- Reduced the size of 12-layer teacher models by 50-80% while having accuracy drops of only 8-10% and performed comparative analysis on different sizes and their accuracies.